

-1.5: 4 problem parts missing

8.5
10

Section 8.1

2. $B(\hat{\theta}) = E(\hat{\theta}) - \theta$

If $\hat{\theta}$ is unbiased, then $B(\hat{\theta}) = 0$ since to be unbiased $\theta = E(\hat{\theta})$

b. $5 = E(\hat{\theta}) - \theta$

$E(\hat{\theta}) = \theta + 5$

3. $B(\hat{\theta}) = E(\hat{\theta}) - \theta$

$= a\theta + b - \theta$

$= \theta(a-1) + b$

* b. Find θ such that $E(\hat{\theta}) = \theta$

This means $E(\hat{\theta}) - \theta = 0 = B(\hat{\theta})$

$\theta(a-1) + b = 0$

$\theta(a-1) = -b$

Not quite. let $\hat{\theta}^* = \frac{\hat{\theta} - b}{a}$.
 $E(\hat{\theta}^*) = \frac{E(\hat{\theta}) - b}{a} = \frac{(a\theta + b) - b}{a} = \theta$.

4. Let $B(\hat{\theta}) = 0$.

Then $MSE(\hat{\theta}) = V(\hat{\theta}) + 0^2$

ii. Let $B(\hat{\theta}) \neq 0$

Then $MSE(\hat{\theta}) = V(\hat{\theta}) + [B(\hat{\theta})]^2$

$MSE(\hat{\theta})$ is the variance plus a penalty term
 $\hookrightarrow$ so $MSE(\hat{\theta}) > V(\hat{\theta})$

6. $E(\hat{\theta}_1) = \theta$

$E(a\hat{\theta}_1 + (1-a)\hat{\theta}_2)$

$= aE(\hat{\theta}_1) + (1-a)E(\hat{\theta}_2)$

$= a\theta + (1-a)\theta$

$= a\theta + \theta - a\theta = \theta$

$\therefore \hat{\theta}_2$ is an unbiased estimator

b. $V(\hat{\theta}_2) = V(a\hat{\theta}_1 + (1-a)\hat{\theta}_2)$

$= a^2 V(\hat{\theta}_1) + (1-2a+a^2) V(\hat{\theta}_2)$

Smallest variance is possible when $a=0$ and $V(\hat{\theta}_1)$ is equal to the variance of one of another point estimator

$\hookrightarrow$ Not quite. See next pg. for a solution.

$$\begin{aligned} \text{bb.) } V(\hat{\theta}_3) &= V[a\hat{\theta}_1 + (1-a)\hat{\theta}_2] = a^2 V(\hat{\theta}_1) + (1-a)^2 V(\hat{\theta}_2) \quad (\text{independence}) \\ &= a^2 \sigma_1^2 + (1-a)^2 \sigma_2^2 \end{aligned}$$

$$\begin{aligned} \frac{dV(\hat{\theta}_3)}{da} &= 2a\sigma_1^2 - 2(1-a)\sigma_2^2 \stackrel{\text{Set}}{=} 0 \Rightarrow 2a(\sigma_1^2 + \sigma_2^2) = 2\sigma_2^2 \\ &\Rightarrow \boxed{a = \sigma_2^2 / (\sigma_1^2 + \sigma_2^2)} \end{aligned}$$

Verifying this is a minimizer: $\frac{d^2 V(\hat{\theta}_3)}{da^2} = 2\sigma_1^2 + 2\sigma_2^2 > 0 \quad \therefore$

$\frac{1}{n} \sum_{i=1}^n Y_i$

$$\frac{1}{n} \sum_{i=1}^n Y_i$$

7.
$$\begin{aligned} V(\hat{\theta}_2) &= V(a\hat{\theta}_1 + (1-a)\hat{\theta}_2) \\ &= a^2 V(\hat{\theta}_1) + (1-2a+a^2) V(\hat{\theta}_2) + \\ &\quad 2a(1-a) \text{Cov}(\hat{\theta}_1, \hat{\theta}_2) \\ &= a^2 V(\hat{\theta}_1) + (1-2a+a^2) V(\hat{\theta}_2) + \\ &\quad (2a-2a^2) \cdot c \end{aligned}$$

Choose $a=1$ to eliminate covariance possibility

8. $Y \sim \exp(\theta)$

$$E(\hat{\theta}_1) = E(Y_1) = \theta$$

$$E(\hat{\theta}_2) = E\left(\frac{Y_1 + Y_2}{2}\right)$$

$$= \frac{1}{2} E(Y_1) + \frac{1}{2} E(Y_2) = \frac{1}{2} \theta + \frac{1}{2} \theta = \theta$$

$$E(\hat{\theta}_3) = E\left(\frac{Y_1 + 2Y_2}{3}\right)$$

$$= \frac{1}{3} E(Y_1) + \frac{2}{3} E(Y_2)$$

$$= \frac{1}{3} \theta + \frac{2}{3} \theta = \theta$$

$$E(\hat{\theta}_5) = E(\bar{Y}) = E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right)$$

$$= \frac{1}{n} \sum_{i=1}^n E(Y_i)$$

$$= \frac{1}{n} \cdot n \theta = \theta$$

a. Everything except order statistic

b. $V(Y_i) = \theta^2$

$$V(Y_1 + Y_2/2) = \frac{1}{4} V(Y_1) + \frac{1}{4} V(Y_2)$$

$$= \frac{1}{4} \theta^2 + \frac{1}{4} \theta^2 = \frac{1}{2} \theta^2$$

$$V(Y_1 + 2Y_2/3) = \frac{1}{9} V(Y_1) + \frac{4}{9} V(Y_2)$$

$$= \frac{1}{9} \theta^2 + \frac{4}{9} \theta^2 = \frac{5}{9} \theta^2$$

$$V\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n^2} \sum_{i=1}^n V(Y_i)$$

$$= \frac{1}{n^2} \cdot n \theta^2 = \frac{\theta^2}{n} = \frac{1}{n} \theta^2$$

c. $\bar{Y}$ has the smallest variance

10. Let $Y = \#$ of breakdowns for car model $Y \sim \text{Poisson}(\lambda)$

a. Sample mean is a good estimator

$$E(\bar{Y}) = E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n} \cdot n \lambda = \lambda$$

↑
✓

$\frac{1}{n} \cdot n \lambda = \lambda$

10b & 10c missing

10c.) An estimator of the form $a\bar{Y} + b\bar{Y}^2$, where a, b are constants, seems like a natural choice.

$$\begin{aligned} E(a\bar{Y} + b\bar{Y}^2) &= aE(\bar{Y}) + bE(\bar{Y}^2) = a\lambda + b\{V(\bar{Y}) + [E(\bar{Y})]^2\} \\ &= a\lambda + b\left[\frac{\lambda}{n} + \lambda^2\right] \\ &= \left(a + \frac{b}{n}\right)\lambda + b\lambda^2 \end{aligned}$$

We want $a + \frac{b}{n} = 4$ & $b = 1$.

Note: $b = 1 \Rightarrow a + \frac{b}{n} = a + \frac{1}{n} = 4 \Rightarrow a = 4 - \frac{1}{n}$.

Hence, $(4 - \frac{1}{n})\bar{Y} + \bar{Y}^2$ is an unbiased estimator for $E(C) = 4\lambda + \lambda^2$.

$$\begin{aligned} 10b.) E(C) &= E(3Y + Y^2) = 3E(Y) + E(Y^2) = 3\lambda + \{V(Y) + [E(Y)]^2\} \\ &= 3\lambda + \lambda + \lambda^2 \\ &= 4\lambda + \lambda^2 \end{aligned}$$

Section 8.5

39 Let $Y \sim \text{Gamma}(2, \beta)$

Let $W = 2Y/\beta$

Recall: $W \sim \chi^2_4$

Find a and b s.t. $P(a \leq W \leq b) = .9$

$$P(W > b) = .05 \quad \text{and} \quad P(W < a) = .05$$

$$b = 9.48773$$

$$a = .710721$$

$$(.7107 \leq 2Y/\beta \leq 9.48773)$$

$$\Rightarrow \left(\frac{1}{9.48773} \leq \beta/2Y \leq \frac{1}{.710721} \right)$$

$$\Rightarrow \left(\frac{2Y}{9.48773} \leq \beta \leq \frac{2Y}{.710721} \right) \checkmark$$

40 Let $Y \sim N(\mu, \sqrt{1})$ $\bar{Y} \sim N(\mu, 1/\sqrt{n})$

Choose $\hat{\theta} = Y$

$$\sigma_{\hat{\theta}} = 1/\sqrt{n}$$

$$95\% \text{ CI } \bar{Y} \pm .025 n^{-1/2}$$

$$\bar{y} \pm 1.96 n^{-1/2}$$

b. $(-\infty, \bar{y} + 1.645 n^{-1/2})$

c. $(\bar{y} - 1.645 n^{-1/2}, \infty)$

b and c missing

41.

$$Y \sim N(0, \sigma^2) \quad Y^2/\sigma^2 \sim \chi^2_1$$

Find a and b s.t. $P(a \leq Y^2/\sigma^2 \leq b) = .95$

This means $P(Y^2/\sigma^2 > b) = .025$ & $P(Y^2/\sigma^2 < a) = .025$

$$b = 5.02389$$

$$a = .0009821$$

$$\Rightarrow (.0009821 \leq Y^2/\sigma^2 \leq 5.02389)$$

$$\Rightarrow \left(\frac{.0009821}{\sigma^2} \leq Y^2 \leq \frac{5.02389}{\sigma^2} \right)$$

4b. $\sum_{i=1}^n Y_i \sim \text{Gamma}(2n, \beta) \equiv \text{Gamma}(\frac{41}{2}, \beta)$

Recall $2Y/\beta \sim \chi^2_n$

$\therefore \frac{1}{\beta} 2 \sum_{i=1}^n Y_i \sim \chi^2_{4n}$ ✓

b. 95% CI: $\left(\frac{2n\bar{Y}}{\chi^2_{0.025}} \leq \beta \leq \frac{2n\bar{Y}}{\chi^2_{0.975}} \right)$ ✓

c. $\Rightarrow \left(\frac{2 \cdot 5 \cdot 5.39}{\chi_{0.025, df=20}} \leq \beta \leq \frac{2 \cdot 5 \cdot 5.39}{\chi_{0.975, df=20}} \right)$

$\Rightarrow (1.577 \leq \beta \leq 5.620)$ ✓

For part (a.), need more explanation as to why W is a pivotal quantity.

-e.g., Since W is a function of the sample measurements and β , where β is the only unknown quantity, and the prob. dist. of W doesn't depend on β , W is a pivotal quantity.